

Project Documentation

Product inventory management

1. Introduction

- ***Project Title: Product inventory management***
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2.Project overview

The Product Inventory Management System is a backend-focused web application developed

using Node.js, Express.js, and MongoDB, designed to provide a secure and scalable solution

for managing product data through RESTful APIs. The system enables users to create

accounts, log in securely, and access protected routes using JWT-based authentication,

ensuring that only authorized users can perform operations. It supports complete CRUD

functionality, allowing users to add new products, view their inventory, update product details

such as price or quantity, and delete items when needed.

3.key Objectives:

- ***Ensure user-specific data isolation so each user can manage only their own products.***
- ***Implement scalable backend architecture using Node.js, Express.js, and MongoDB.***

4.Target Users:

- ***Small business owners managing product stock and inventory records.***
- ***Freelancers or sellers tracking items or services in a structured way.***
- ***Developers learning backend API development and authentication workflows.***
- ***Students working on backend or full-stack development projects.***
- ***Anyone needing a secure and simple backend system for managing product data.***

5.Project instruction

- ***Create backend project folder***
- ***Configure server using Express.js***
- ***Setup MongoDB database connection***
- ***Implement authentication APIs***
- ***Implement product CRUD APIs***
- ***Test APIs using Thunder Client in VS Code***

6. System Features

The system provides the following features:

Add new products

View product list

Update product information

Delete products ***System Architecture***

7.System Architecture

The system follows a three-layer architecture:

1) Client Layer

***** User sends requests to the server.***

2) Application Layer

***** The backend server processes requests using APIs built Node.js and Express.js.***

3) Database Layer

***** Product data is stored and retrieved from Mongo db***

8. Database Structure

Example Product Collection

Product

_id

name

price

quantity

9. ER Diagram (Concept)

Entities:

Product

ProductID

ProductName

Price

Quantity

Description

Relationship:

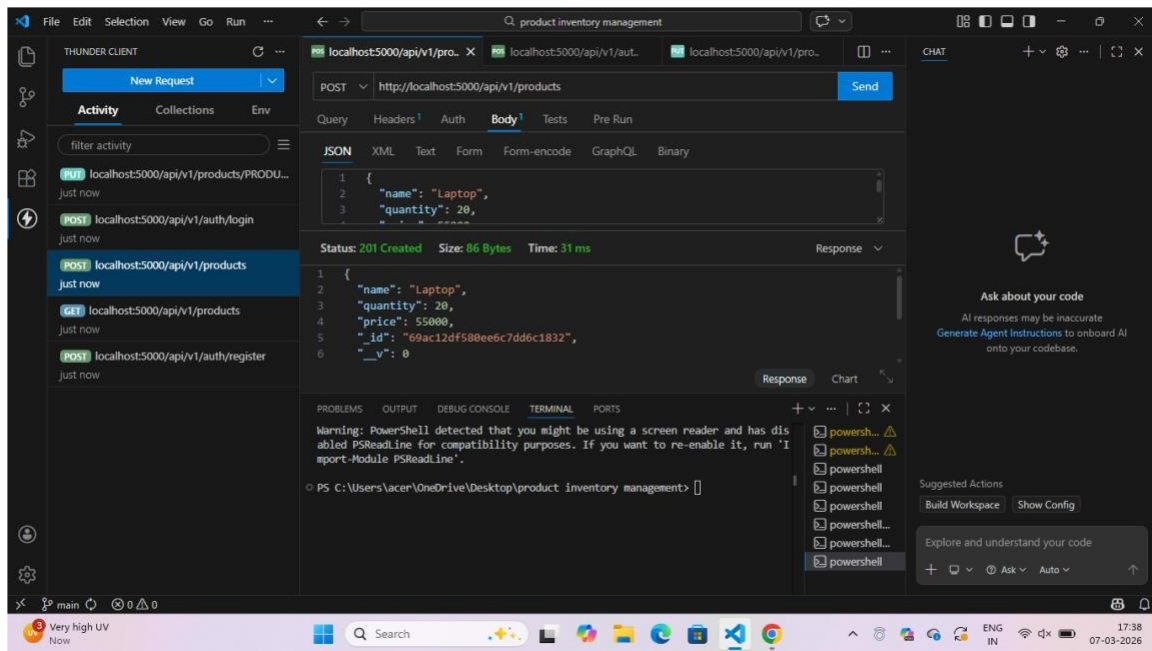
Inventory manages the product details and stock information.

10.API testing

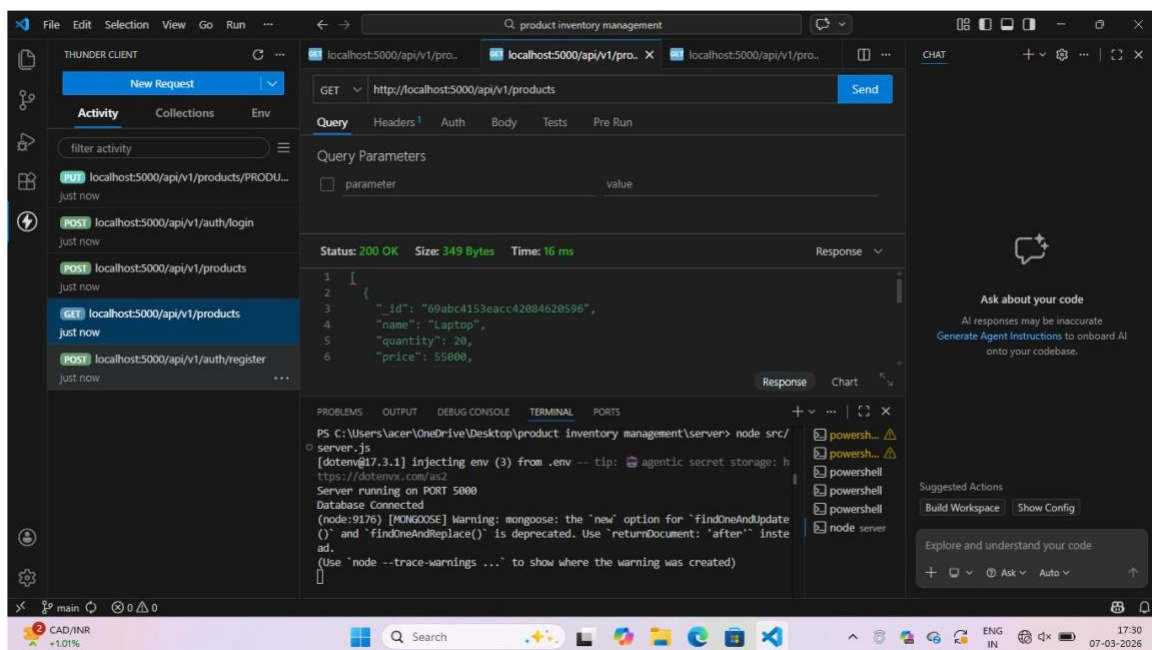
Use thunder client to test Api's

Test API'S

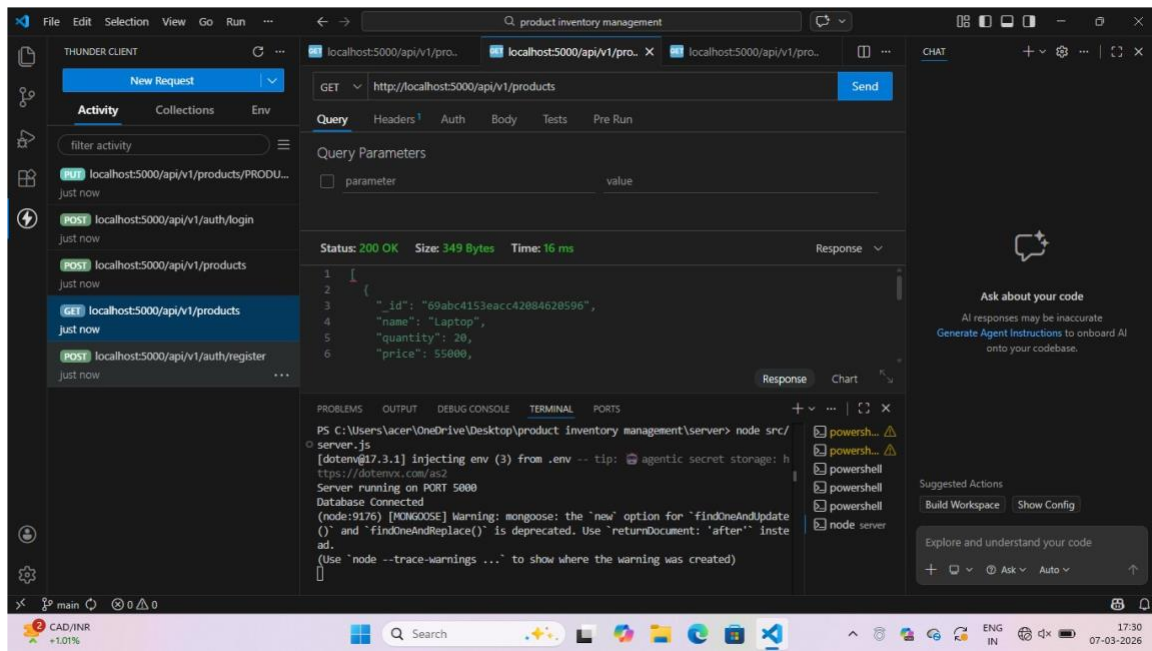
Register user



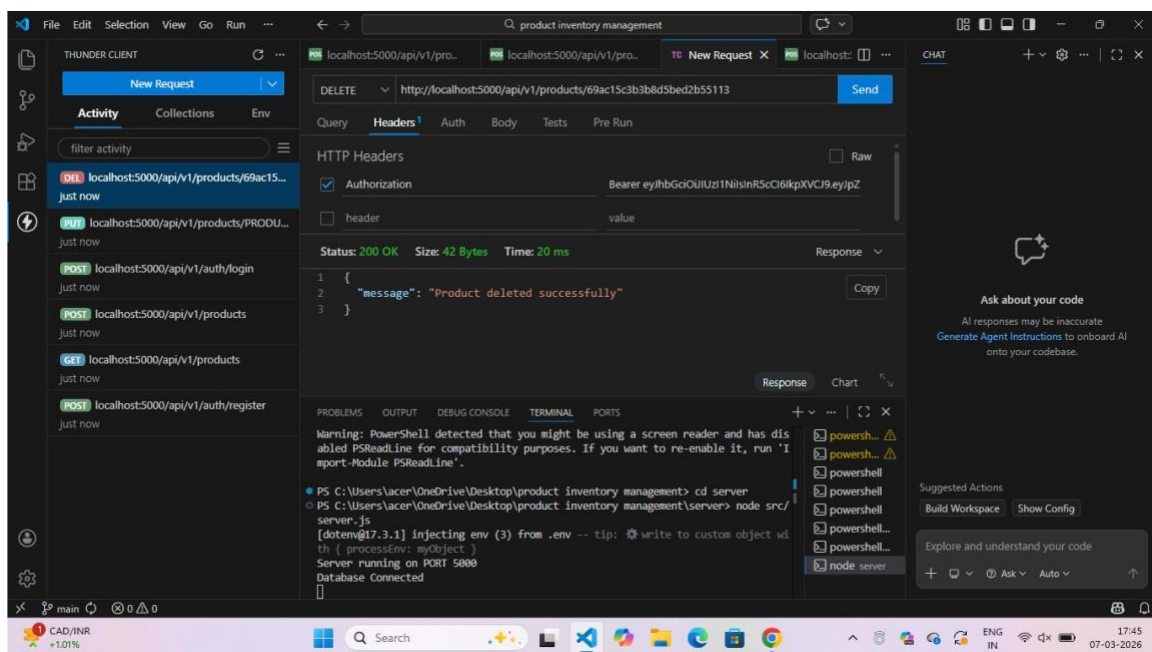
Read Product



Update Product



Delete Product



11. Advantages

The system provides several advantages:

- * Easy product management*
- * Accurate inventory tracking*
- * Reduces manual record keeping*
- * Improves efficiency in inventory operations*

**** Faster access to product data***

12. Limitations

No user authentication module

Basic inventory tracking only

These features can be added in future versions.

13. Future Enhancements

Future improvements may include:

- i. User authorization and Dashboard with analytics***
- ii. Low stock alerts***
- iii. Barcode scanning integration***
- iv. Frontend user interface***
- v. Integration with sales systems***

14. Conclusion

The Product Inventory Management System provides an efficient solution for managing product inventory digitally. By using modern technologies such as Node.js, Express.js, and MongoDB, the system offers reliable and scalable inventory management.

This system helps businesses maintain accurate product records, improve efficiency, and reduce manual workload.

